CALCULUS · UNIT 6: THE FUNDAMENTAL THEOREM OF CALCULUS

Coffee Shop Rush

Use the Fundamental Theorem of Calculus to track customers, inventory, and wait times at a coffee shop.

CP1

2–3 class periods

Name: _____ Date: _____ Period: _____

The Scenario

The Daily Grind coffee shop gets slammed from 6 to 10 a.m. Customers arrive at a changing rate, baristas serve at a steady rate, and the line grows and shrinks.

You will use the Fundamental Theorem of Calculus to track the line, the milk supply, and the average wait, and then recommend how many baristas to schedule.



Arrivals

$A(t) = 60 + 40\sin(\pi t/4)$

customers per hour

Service

Each barista serves

30 customers per hour

Learning Goals

- I can use the FTC to compute net change and accumulation.
- I can analyze accumulation functions defined by integrals.
- I can find average value and interpret it.

Project Timeline

When	What you will do
Day 1	Line Length
Day 2	Milk & FTC
Day 3	Staffing
Final	Rush-Hour Staffing Plan + reflection

Part 1: Line Length

Day 1 · With 3 baristas (90 per hour) the early line clears quickly. A new line starts when arrivals pass 90 per hour at t ≈ 1.08 (7:05 a.m.), so L(t) = ∫ from 1.08 to t of (A(x) − 90) dx.

Shop Data Cards (t = hours after 6 a.m., 0 ≤ t ≤ 4)

Arrivals	Service	Line at 6 a.m.
A(t) = 60 + 40 sin(πt/4)	S = 30 per barista per hour	5 people

Milk Use	Milk Stock	Wait Rule
M(t) = 3 + 2t gallons per hour	12 gallons at 6 a.m.	Line should stay under 25 people

1. Find L′(t) using FTC Part 1. When is the line growing?

2. When is the line longest? Justify.

3. Find the maximum line length (technology allowed).

4. Find L(4) and interpret.

Part 2: Milk & FTC

Day 2 · M(t) = 3 + 2t gallons per hour; 12 gallons at 6 a.m.

5. Write the milk remaining as G(t) = 12 − $\int_0^t$ M(x) dx and find G(2).

6. Find $\frac{d}{dt}$ ∫₀^(t²) (3 + 2x) dx.

7. A delivery adds 20 gallons at 8 a.m. Write the milk remaining for t > 2 and find when it runs out.

8. Find the average value of A(t) on [0, 4].

Part 3: Staffing

Day 3 · Wait rule: keep the line under 25 people.

9. With 2 baristas (60/hour), write $L_2(t) = 5 + \int_0^t (A - 60)\,dx$ and find $L_2(4)$.

10. With 3 baristas, show the line never exceeds 25.

11. Write the staffing plan with the number of baristas by hour and the milk order.

Final Product: Rush-Hour Staffing Plan

Your final product must include:

- Line model with FTC Part 1
- Maximum line and timing
- Milk accumulation with delivery
- Average arrival rate
- Staffing plan

Reflection

R1. How does FTC Part 1 let you find when the line is longest without integrating?

R2. What assumptions in the model break down, and why?

Scoring Rubric

Criteria	4 – Exceeds	3 – Meets	2 – Approaching	1 – Beginning
Mathematical Accuracy	All calculations and solutions are correct; answers include units and are checked.	Nearly all work is correct; minor errors do not change conclusions.	Several errors; some conclusions are affected.	Many errors or missing work.
Reasoning & Justification	Clear, complete explanations connect every answer to the situation; uses precise vocabulary.	Explanations are mostly clear and connected to the situation.	Explanations are brief or partly unclear.	Little or no explanation.
Application & Design	Integrals of rates, accumulation functions, and average values are correct and support the staffing decision. Goes beyond the minimum with a creative, realistic choice.	Integrals of rates, accumulation functions, and average values are correct and support the staffing decision.	Design is incomplete or only partly uses the math from the unit.	Design is missing or unrelated to the unit math.
Communication & Presentation	Plan includes models, FTC work, graphs, and a clear staffing schedule. Polished, organized, and visually engaging.	Plan includes models, FTC work, graphs, and a clear staffing schedule. Organized and easy to follow.	Product is missing required parts or is hard to follow.	Product is mostly missing.